

Optimisation Methods for Antenna Arrays for Wireless Power Transfer

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Abstract—Wireless power transfer techniques using field synthesis and optimization is studied; the system is implemented via an array of inset-patch antennas at 5.8 GHz. Optimisation techniques are presented and compared in order to maximize end-to-end power transfer efficiency (PTE) and field focality. For a single R_x element positioned 1λ away in the near-field, a noticeable improvement of a PTE is observed. A larger array with more elements allows for focusing over greater distances, as well as potentially multiple R_x targets at different locations.

Index Terms—Arrays, patch antennas, focusing, field synthesis, wireless power transfer, convex optimisation

I. INTRODUCTION

WIRELESS power transfer (WPT) is useful for powering devices at a distance where mobility is important, or if the device is in a difficult-to-reach location. The convenience afforded by WPT technology has therefore made it a major area of research interest. Different mechanisms for achieving WPT are viable depending on the frequency, distance, power and efficiency requirements; inductive coupling is the most common commercially. Using focused antenna arrays to achieve WPT in the radiative near field and far field is also possible, especially for powering devices at a greater distance (e.g. 10 - 100 cm).

The goals of WPT are usually: to maximize transmission distance, transmission efficiency and transmitted power. Typically end-to-end PTE is the metric which most optimizations are concerned with. The quantity of transmitted power varies based on the application; as little as a few milliwatts (mW) can be used for remote sensors, whereas larger devices such as smartphones and laptop computers require 1-2 W. Distance requirements are also highly variable depending on the application.

Two different methods of field focusing are presented in the following sections. In most cases, consideration of the scattering parameters is required to maximize PTE. In scenarios where the S-matrix between the T_x and R_x elements are not known or is variable (e.g. in the case of charging multiple different devices), field focality becomes the important consideration for WPT.

II. FOCUSING WITH MICROSTRIP PATCH ARRAYS

Microstrip patch antennas are easily manufactured, physically robust, low-profile with moderate gain. A square array of patches can be used for WPT; by adjusting the phase and current on the patches, field characteristics can be adjusted to

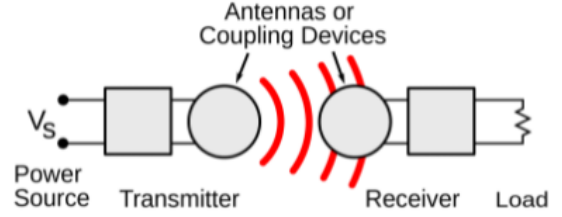


Fig. 1. Schematic representation of basic WPT setup, without intervening elements between T_x and R_x element. Diagram from Wikipedia, Wikimedia Foundation.

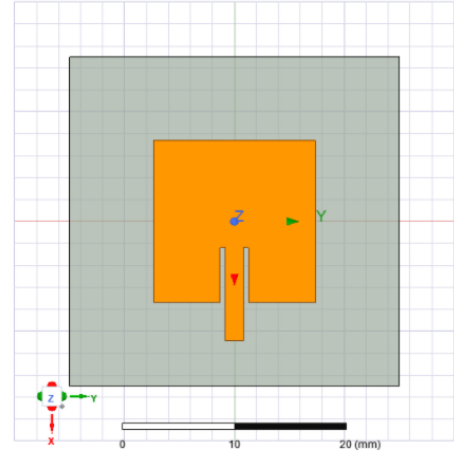


Fig. 2. View of inset-patch antenna designed in Ansys HFSS (orange, PEC), mounted on Rogers RO3003 substrate (light green, $\epsilon_r = 3.0$). Dimensions and specifications are listed in table I.

maximize PTE and/or field focality [1]. The patch is optimized for 5.8 GHz, a common band for medical/scientific devices [2], and mounted on a 0.4 mm layer of Rogers RO3003 substrate (fig. 2).

The system is simulated using Ansys HFSS (2021 R1). In this project a 3×3 element array excited by 9 lumped ports is used for simplicity and to reduce computation time, although in practice an even number of elements is most ideal as the feed network would be symmetric. The target for charging is positioned at one wavelength ($\lambda = 5.17$ cm) away from the array, in the near field, using an identical patch for the R_x element; greater distances can also be considered for a larger array. The PTE is determined by integrating the magnitude of the Poynting vector over the ports. All ports are assigned a

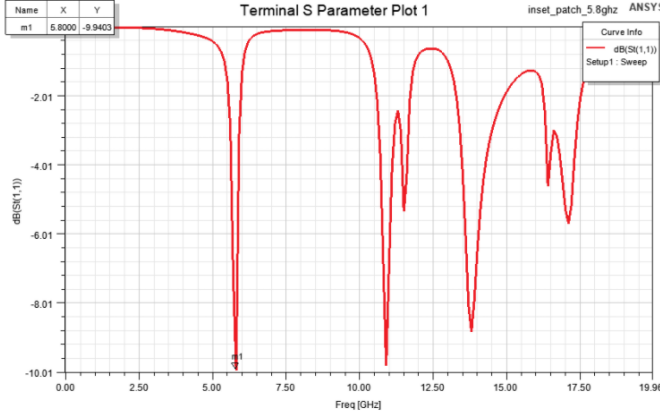


Fig. 3. S_{11} frequency sweep for patch in fig. 2. At 5.8 GHz, the design frequency, $S_{11} = -9.94$ dB.

TABLE I
DESIGN PARAMETERS OF INSET-FEED MICROSTRIP PATCH AT 5.8 GHz
(ALL UNITS IN MM).

Patch width/length	14.75
Feed line width	1.70
Feed line length	3.70
Inset gap width	2.55
Inset gap length	5.00

characteristic impedance of 50Ω .

For convenience the center element is labelled (0,0), with other elements labelled ($\{0, \pm 1\}, \{\pm 1, \pm 1\}$) accordingly. Spacing between elements is set to $\lambda/2$ to minimize element-element coupling.

III. UNFOCUSED ARRAY

For a totally unfocused array, the 3×3 array is uniformly excited with 1 W at each port. This scenario is used as a benchmark case against which the other examples are compared. Due to the R_x element being placed relatively close (1λ), the baseline PTE is higher than would be expected in a more general application, at $\eta = 0.157$.

IV. CONJUGATE-PHASE (CP) FOCUSING

To improve the PTE, one approach is to adjust the phase of the elements such that the wavefronts constructively interfere at the focal point (i.e. the location of the R_x element), as shown in fig. 4 [1]. This has the benefit of only needing to know the displacement between the T_x and R_x elements to be implemented. The accumulated phase between the R_x patch and any array element can be calculated by:

$$\phi = \frac{2\pi}{\lambda} \|r_{\text{focal}} - r_{\text{elem}}\|_2 \quad (1)$$

The assigned phases are listed in table II; power at each port is an uniform 1 W. The resulting PTE is $\eta = 0.175$, showing improvement over the unfocused array.

TABLE II
RELATIVE ELEMENT PHASES (RAD) FOR CP-FOCUSING OF ARRAY.

Element	Phase (rad)
(0,0)	0.0
(0, ± 1), (± 1 ,0)	0.74163
(± 1 , ± 1)	1.4121

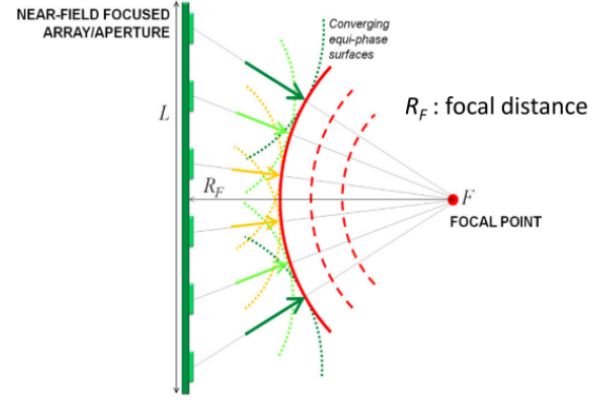


Fig. 4. Diagram illustrating principle of conjugate-phase (CP) focusing in the near field. Equi-plane surfaces are made to converge at the designed focal point for increased PTE [1].

V. METHOD BY MINIMIZING SIDELOBES

The element weights and phases can both be adjusted in order to selectively focus the E-field over some target region for WPT, while minimizing unwanted fields elsewhere. In general the problem can be stated as a least squares problem:

$$\arg\min_{\mathbf{w}} \|E_{\text{ref}} - E_{\mathbf{w}}\|_2 \quad (2)$$

Where E_{ref} is the target E-field distribution and $E_{\mathbf{w}}$ is the produced E-field, dependent on the vector of element weights $\mathbf{w}$. A typical constraint for WPT is power, although increasing or decreasing available power does not affect PTE. Sidelobe levels can also be a constraint, as described in [3]. The optimization problem is written as:

$$\begin{aligned} &\max \text{Re}\{E_{\text{NF}}(0, 0, z_o)\} \\ &\text{s.t. } |E_{\text{NF}}(x, y, z_o)| < \rho, \quad (x, y) \in \Omega_{\text{SLL}} \end{aligned}$$

To ensure minimal field leakage outside the target region, the problem can be solved for $\rho = 0$; the field within the focal region (defined as the area of the R_x patch) is expressed in the basis of the fields produced by each element of the T_x array. The resulting element excitations are listed in table III. The PTE is $\eta = 0.161$, showing a slight improvement over the unfocused array. Similar to the CP-phase method, this depends solely on the system's geometry and does not require any knowledge of the R_x device.

VI. METHOD BY USING S-MATRIX

A method of optimizing for PTE is presented in [4] using the S-matrix of the system. The optimization problem can be formulated as [5]:

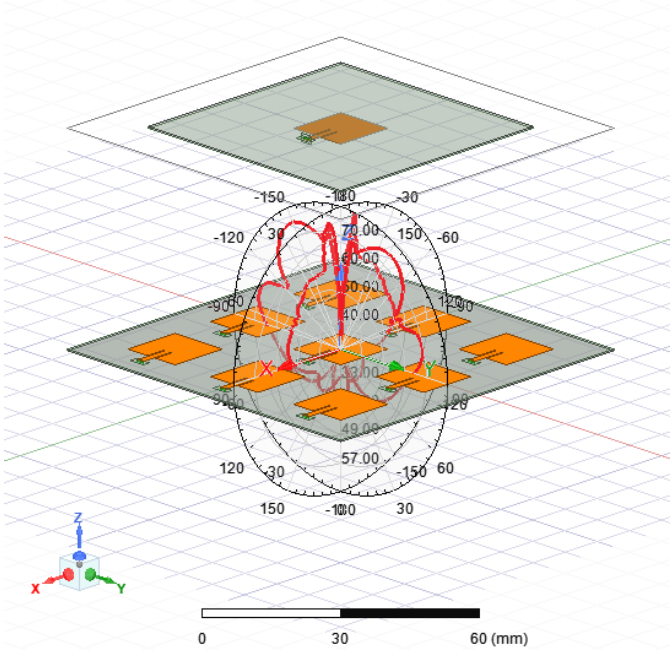


Fig. 5. Dimetric view of WPT system setup in HFSS, show 3×3 element T_x array. An identical single patch is used as the R_x element. The near-field Poynting vector in the E and H-planes (dB) are shown for the CP-focusing case. The substrate thickness is 0.4 mm.

TABLE III

RELATIVE ELEMENT MAGNITUDE AND PHASES (RAD) FOR FOCUSING VIA SIDELobe MINIMIZATION.

Element	Magnitude, Phase (rad)
(0,0)	0.76150, -1.0702
else	1.0, 0

$$\max \frac{\mathbf{x} \mathbf{A}^T \mathbf{x}}{\mathbf{x}^T \mathbf{x}}$$

$$\mathbf{A} = [S_{rt}]^T [S_{rt}]$$

Where $\mathbf{x}$ is the vector of excitations corresponding to the array elements. S is the scattering matrix, with subscripts r and t denoting the R_x and T_x elements respectively:

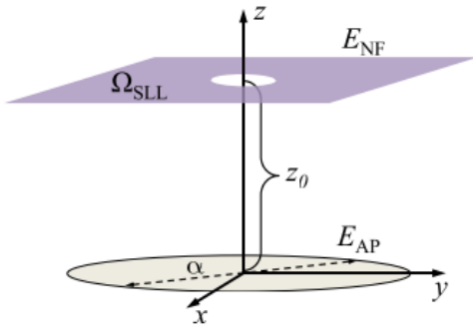


Fig. 6. Diagram of focusing setup from [3]. The focal plane E_{NF} show in purple describes a pre-defined maximum sidelobe level Ω_{SLL} outside the target region. The focal distance is fixed at z_0 .

TABLE IV
RELATIVE ELEMENT MAGNITUDE AND PHASES (RAD) FOR OPTIMIZING PTE USING S-MATRIX.

Element	Magnitude, Phase (rad)
(0,0)	0.52969, 1.0040
(0, ± 1), (± 1 ,0)	0.33801, 0.52689
(± 1 , ± 1)	0.24048, 0

TABLE V
COMPARISON OF END-TO-END PTE FOR WPT METHODS, 1.0 BEING 100% EFFICIENT.

Unfocused array	0.157
CP focusing	0.175
Sidelobe minimization	0.161
S-matrix method	0.161

$$S = \begin{bmatrix} [S_{tt}] & [S_{tr}] \\ [S_{rt}] & [S_{rr}] \end{bmatrix} \quad (3)$$

The disadvantage here is that without knowledge of the S-matrix between the array and R_x device beforehand, the method cannot be implemented.

For multiple R_x elements, this method is used under the additional constraint that the power received at each R_x element is equal ("flat-top" condition [4]). In the case of a single R_x element here, introducing additional constraints is unnecessary. Therefore, the problem becomes an unconstrained optimization. The resulting element excitations are listed in table IV. The PTE is $\eta = 0.161$, the same as the example using sidelobe minimization, showing a slight improvement over the unfocused array.

VII. CONCLUSION

A patch antenna array is presented for near-field WPT. Two different optimization methods as well as CP focusing are compared against an unfocused and array. The improvement in PTE is noticeable and demonstrates the their viability for enhancing WPT; larger arrays will allow for WPT over greater distances. Using the techniques shown it is also possible to selectively charge multiple devices at once, by appropriately choosing weights for the array elements.

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